

- **Slide 1: Title & Project Metadata:** Introduce Satyanaam-Food, a food ordering web application developed for the 2025-2026 session under Mr. Kailash Purbia.
- **Slide 2: The Problem Statement:** Address human errors in phone ordering and high commissions from third-party apps by providing a direct, commission-free platform.
- **Slide 3: The Technology Stack:** Utilize a modified MERN stack with Vanilla JavaScript, Bootstrap 5, Axios, Node.js, Express, MongoDB, and Redis for a lightweight, fast application.
- **Slide 4: System Architecture & Workflow:** Optimize performance using a Redis caching layer to serve menu data instantly and reduce database load on MongoDB.
- **Slide 5: Database Schema Design & Security:** Implement secure data handling with bcrypt password hashing, JWT for sessions, and Role-Based Access Control for menu modifications.
- **Slide 6: Core Features Walkthrough:** Provide a dynamic single-page experience with backend-calculated totals, automated email receipts via Nodemailer, and an Admin Dashboard.
- **Slide 7: Software Testing Strategy:** Ensure reliability through Unit Testing for authentication, White Box testing for APIs, and Black Box testing for the customer checkout journey.
- **Slide 8: Future Scope & Conclusion:** Conclude the project as a high-performance system with future plans for Razorpay integration, WebSockets for tracking, and inventory management.

Slide 1: Title & Project Metadata

- **What to Say:** "Good morning, examiners and teachers. Today, I am presenting my final year project, **Satyanaam-Food**, which is a food ordering web application. This project was developed during the 2025-2026 session under the guidance of Mr. Kailash Purbia."

Slide 2: The Problem Statement

- **What to Say:** "Traditional restaurants face a lot of issues today. Taking orders over the phone leads to human errors and missed calls during busy hours. On the other hand, third-party apps like Swiggy and Zomato take a massive 15% to 30% commission, which ruins restaurant profits. **Satyanaam-Food** solves this by giving the restaurant its own direct, commission-free digital platform."

Slide 3: The Technology Stack

- **What to Say:** "For this project, I used a modified MERN stack. On the frontend, instead of using a heavy framework like React, I used **Vanilla JavaScript, Bootstrap 5, and Axios**. This keeps the application incredibly lightweight and fast. The backend runs on **Node.js and Express** with a **MongoDB** database, and I added a **Redis** caching layer to speed up menu loading times."

Slide 4: System Architecture & Workflow

- **What to Say:** "This slide shows how data flows through the application. When a customer opens the website to browse the menu, the backend checks the **Redis cache** first. If the menu is there, it serves it instantly. If it is a cache miss, it pulls the data from MongoDB and updates the cache for the next user. This structure drastically reduces database load."

Slide 5: Database Schema Design & Security

- **What to Say:** "Our database uses three main collections: **Users, Menus, and Orders**. Security is integrated directly into the design. Passwords are never stored in plain text; they are safely hashed using **bcrypt**. User sessions are handled securely using **JSON Web Tokens (JWT)**, and the system uses Role-Based Access Control to ensure only admins can modify the menu."

Slide 6: Core Features Walkthrough / Live Demo

- **What to Say:** "Looking at the user interface, customers get a clean, single-page shopping experience. They can search by category and add items to their cart dynamically without page reloads. Order totals are calculated strictly on the backend to prevent tampering. Once an order is placed, **Nodemailer** sends an automated email receipt to the customer, and the order pops up instantly on the Admin Dashboard."

Slide 7: Software Testing Strategy

- **What to Say:** "To make sure the application works perfectly, I conducted multiple layers of testing. I used **Unit Testing** to verify our JWT authentication middleware. I used **White Box testing** via Postman to test our backend API endpoints directly, and **Black Box testing** to ensure the entire customer checkout journey works flawlessly on the frontend."

Slide 8: Future Scope & Conclusion

- **What to Say:** "In conclusion, Satyanaam-Food successfully creates a high-performance, independent e-commerce system for a modern restaurant. In the future, I plan to expand this project by adding a secure digital payment gateway like Razorpay, implementing real-time order tracking using WebSockets, and creating an automated inventory management system. Thank you, and I am now ready for your questions."